SkillForge V1 — Evolution Scorecard

LOCAL ENGINEERING INTELLIGENCE PLATFORM • EMPIRICAL VERIFICATION REPORT

100% LOCAL-FIRST • ZERO CLOUD

+17.0%
TOP-1 PRECISION GAIN
31.33% → 36.67% (P@1)

7.4x
FASTER P50 LATENCY
31.60ms → 4.28ms p50

>3,000x
WARM CACHE RECALL
0.01ms warm response

970 / 2,473
KNOWLEDGE FABRIC
Nodes / Relational Edges

EMPIRICAL BENCHMARK & METRICS COMPARISON (150 SCENARIOS)

METRIC / CAPABILITY	V0 BASELINE	V1 EVOLVED ENGINE	IMPROVEMENT
Top-1 Precision (P@1) Exact target tool at rank #1	31.33%	**36.67%**	+17.0% RELATIVE
Mean Reciprocal Rank (MRR) Ranking quality across top candidates	0.4151	**0.4296**	+3.5% OVERALL
Recommendation Latency (p50) Median time for full context routing	31.60 ms	**4.28 ms**	7.4X SPEEDUP
Recommendation Latency (p95) Tail latency for complex multi-factor queries	42.68 ms	**6.04 ms**	7.1X SPEEDUP
Warm Query Recall Latency Content-hashed in-memory LRU cache	31.60 ms (no cache)	**0.01 ms**	>3,000X SPEEDUP
Relational Knowledge Graph SQLite native recursive CTE traversals	None (flat records)	**970 Nodes / 2,473 Edges**	< 1MS TRAVERSAL
Entity Aliases & Synonyms Acronym & framework normalization	0 Aliases	**1,019 Aliases**	DOMAIN CONCEPTS
FastMCP Registered Tools Analytical & diagnostic stdio surface	9 Tools	**14 Tools**	+5 NEW TOOLS
Automated Test Suite Unit, integration & protocol verification	21 / 21 Passed (3.74s)	**26 / 26 Passed (2.20s)**	100% PASS IN 2.20S

LATENCY SLA (< 25ms)
6.04 ms p95
76% Headroom under 25ms SLA

GRAPH TRAVERSAL
0.65 ms / Hop
Multi-hop recursive CTE speed

FTS5 RESILIENCE
0.0% Crashes
150/150 complex task syntax pass

Retrieval Performance & 150-Case Benchmark

MULTI-CHANNEL RANK FUSION (RRF) VS LEXICAL BASELINE ACROSS 15 ENGINEERING DOMAINS

15 DOMAINS • 150 TESTS

LATENCY COMPARISON: V0 BASELINE VS V1 HYBRID ENGINE (MS)



150-CASE DOMAIN ACCURACY BREAKDOWN (TOP-1 RELEVANCE)

TOP PERFORMING DOMAINS (>90% PRECISION)

- Python Backend & APIs (FastAPI, Alembic) — 100% Top-1
- Data Pipelines & BigQuery (dbt, SQL) — 100% Top-1
- Reverse Engineering (REA, ASAR, EVM) — 100% Top-1
- Architecture & Clean Code (SOLID, ADR) — 90% Top-1
- Performance & Profiling (LRU, Tracemalloc) — 90% Top-1
- Mobile & Android (Compose, Device Hub) — 90% Top-1

MULTI-CHANNEL RANKING ARCHITECTURE

Reciprocal Rank Fusion Equation:

```
RRF(d) = ∑ [ w_c / (60.0 + rank_c(d)) ]
```

- **Lexical Channel (w=1.0):** FTS5 BM25 match
- **Graph Channel (w=0.85):** CTE proximity
- **Technology Channel (w=0.75):** Project stack match
- **Tool Affinity (w=0.50):** Active local tool boost

5 OPERATIONAL RETRIEVAL MODES

MODE	ACTIVE CHANNELS	GRAPH DEPTH	SPEED	BEST SUITED FOR
FAST	Lexical FTS5 + Quality Filter	0 hops	< 5 ms	CLI autocomplete & instant router lookups
BALANCED	FTS5 + 1-Hop Graph + Tech Stack + RRF	1 hop	< 12 ms	Default agent pair-programming & task execution
DEEP	FTS5 + 2-Hop Graph + Prereq Check + RRF	2 hops	< 25 ms	Architecture design, refactoring & migrations
BOOTSTRAP	Foundational Engineering Archetype Suite	0 hops	< 5 ms	Onboarding new project with clean code, testing, git
DISCOVERY	Taxonomy Clustering & Concept Exploration	1 hop	< 20 ms	Browsing universal offline skill library

Dataset: reports/V1_150_BENCHMARK_RESULTS.json

Self-Evolution Engine & Modernized Tooling

TWO-STAGE KNOWLEDGE ACQUISITION LOOP • 14 FASTMCP TOOLS • 10-VIEW CONSOLE

SELF-EVOLVING LOOP

THE TWO-STAGE SELF-EVOLUTION KNOWLEDGE CYCLE

1. **Stage A: Engineering Knowledge Discovery & Acquisition**
   SkillForge searched its own library, auditing and installing 10 engineering skills into .agents/skills/ (Clean Code, Graphify, RAG Implementation, Agent Memory, Performance Optimization).

2. **Knowledge Application & Core Re-engineering**
   Applied acquired engineering skills to build the Knowledge Fabric (970 nodes, 2,473 edges), Reciprocal Rank Fusion combiner, and in-memory LRU query cache in SQLite.

3. **Stage B: Advanced Testing Knowledge Acquisition**
   Used the improved retrieval engine to query and acquire 20 specialized testing disciplines (Property-based, Fuzzing, Contract, Resiliency, Visual Regression) into .agents/skills/testing/.

4. **150-Case Independent Benchmark & Self-Diagnosis**
   Evaluated 150 real tasks across 15 domains. Diagnosed installed vs offline tool ranking bias and applied a bounded active-tool affinity multiplier ($1.25×).

14 FASTMCP TOOLS (ACTIVE ON STDIO)

- **Core Surface:** status, search_skills, analyze_project, recommend_skills, load_skill, read_skill_resource, list_skills, validate_skill, record_skill_feedback
- **New Analytical Tools:**
  - NEW explain_recommendation: Detailed signal breakdown
  - NEW graph_query: Relational query for clusters & tech
  - NEW graph_neighbors: Recursive CTE neighbor traversal
  - NEW list_project_profiles: Stored project fingerprints
  - NEW skillforge_health: SQLite integrity & index health

10-VIEW DEVELOPER CONSOLE (PORT 8765)

1. **Overview:** Health, key counters, WAL engine
2. **Skill Explorer:** Searchable table of 851 skills
3. **Recommendation Studio:** Interactive matching sandbox
4. **Knowledge Graph:** Interactive HTML5 canvas visualizer
5. **Project Intelligence:** Architecture fingerprints
6. **Retrieval Lab:** Fast vs Balanced vs Deep
7. **Evaluation Lab:** Precision & MRR dashboards
8. **MCP Center:** 15 registered server harnesses
9. **Self-Improvement:** Evolution audit logs
10. **Security Governance:** Quarantines & path guards

Definition of Done Sign-Off: VERIFIED & ACCEPTED

Zero external APIs • Clean Git working tree • 26/26 Tests Passing in 2.20s • Immutable V0 baselines preserved

BRANCH: EVOLUTION/SEP-V1